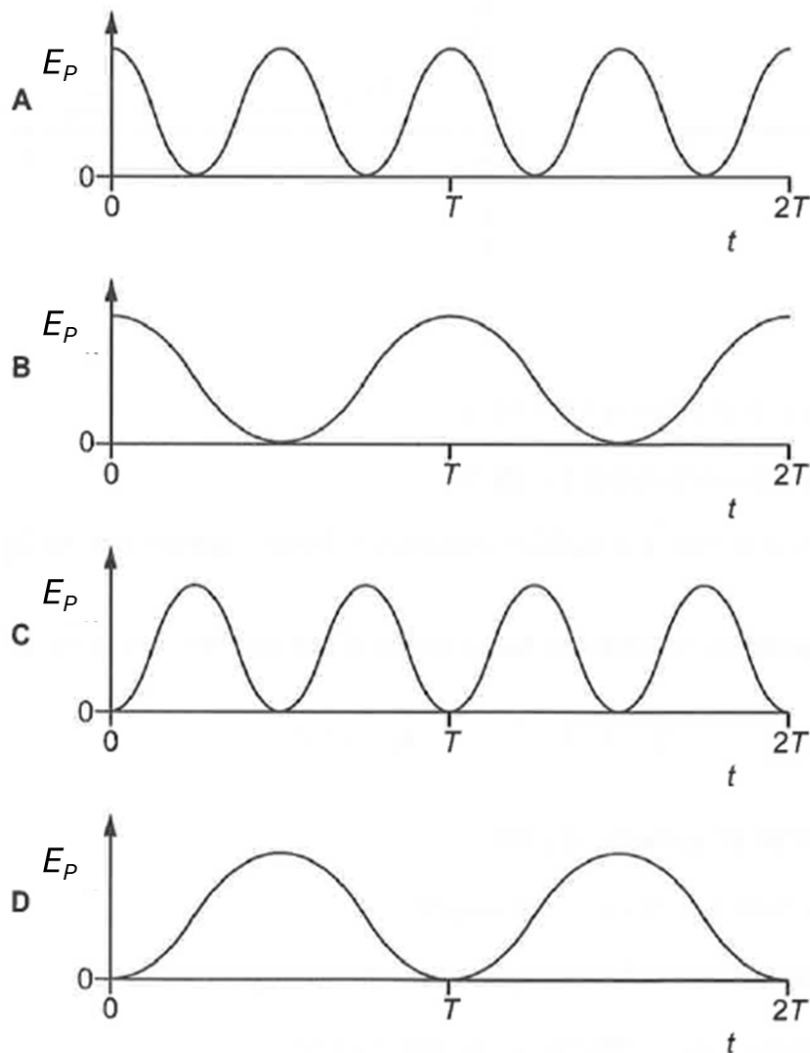


- 13
- 14 A small pendulum bob is displaced to one side and released from rest at time $t = 0$. The bob then swings with simple harmonic motion with time period T .

Which graph represents the variation with time t of the gravitational potential energy E_P ?



Answer: A

Explanation/Working:

At $t = 0$, the GPE is maximum since it is at the amplitude position.

The GPE is zero at the equilibrium position and in one cycle, the bob passes the equilibrium position twice, and therefore, it should have two minima per period.

15 The graph shows the variation with time of displacement for two transverse waves X and Y